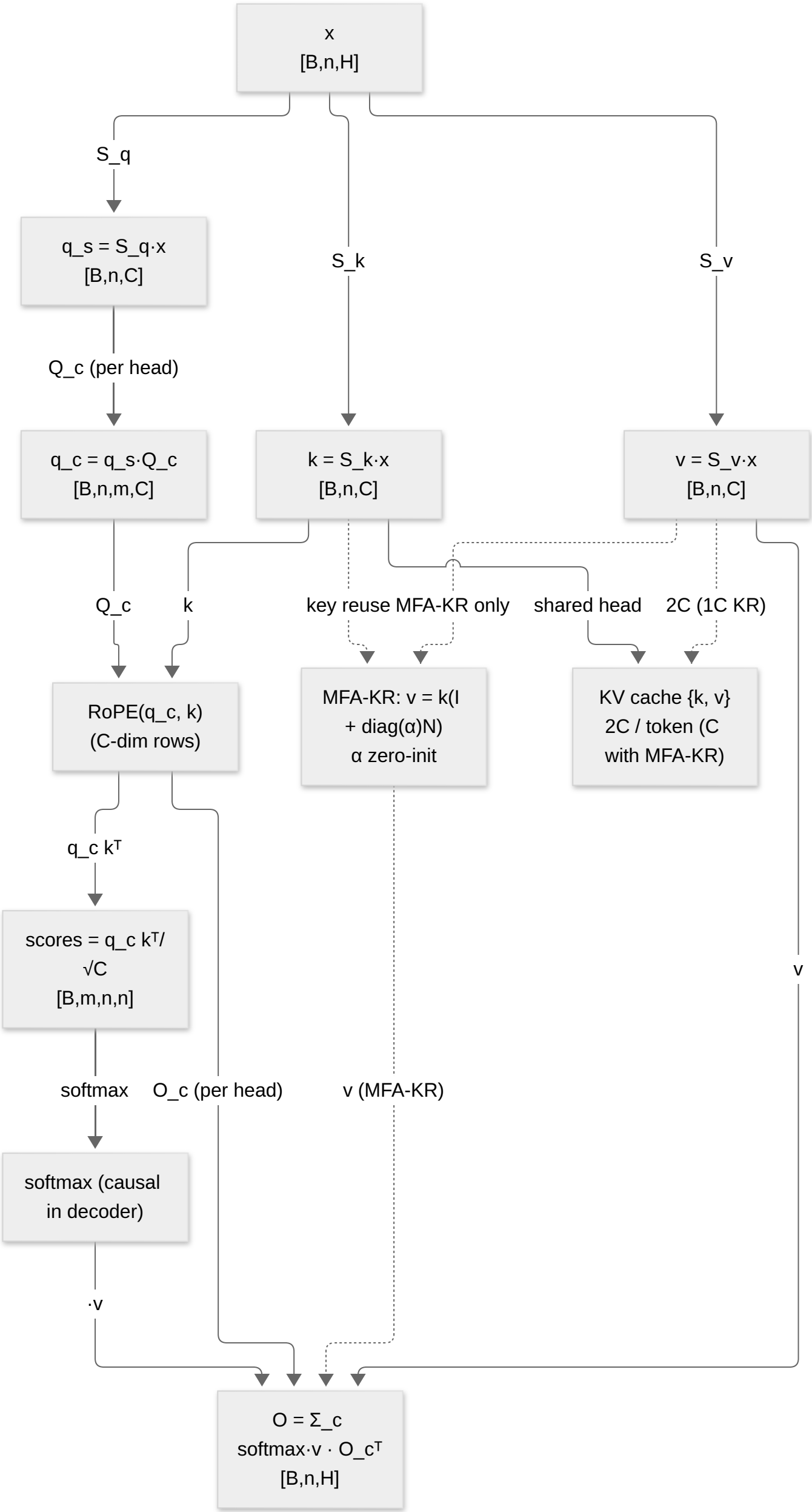

x
[B,n,H]
S_q
S_k
S_v
q_s = S_q·x
[B,n,C]
Q_c (per head)
q_c = q_s·Q_c
[B,n,m,C]
k = S_k·x
[B,n,C]
v = S_v·x
[B,n,C]
Q_c
k
key reuse MFA-KR only
shared head
2C (1C KR)
RoPE(q_c, k)
(C-dim rows)
MFA-KR: v = k(I
+ diag(α)N)
α zero-init
KV cache {k, v}
2C / token (C
with MFA-KR)
q_c k^T
scores = q_c k^T/
√C
[B,m,n,n]
softmax
O_c (per head)
v (MFA-KR)
v
softmax (causal
in decoder)
·v
O = Σ_c
softmax·v · O_c^T
[B,n,H]